

Decent Coaching Classes

VIII SB

Worksheet

Chapters 1 to 8 | Maths Part I | Maharashtra State Board
Draw neat figures wherever required. Show all steps of your working.

Chapter 1 : Rational and Irrational Numbers

Q1. Choose the correct alternative.

- (1) Which of the following is an irrational number?
(A) $22/7$ (B) $\sqrt{2}$ (C) 0.75 (D) -5
- (2) The decimal form of $7/4$ is
(A) 1.25 (B) 1.75 (C) 1.5 (D) 1.7
- (3) Which of the following is smaller? $-3/4$ or $-2/3$
(A) $-3/4$ (B) $-2/3$ (C) both are equal (D) cannot be compared
- (4) The decimal form of $1/3$ is
(A) terminating (B) non-terminating recurring (C) non-terminating non-recurring
(D) none of these
- (5) To show $\sqrt{2}$ on a number line, we draw a right-angled triangle whose two perpendicular sides are each of length
(A) 1 unit (B) 2 units (C) $\sqrt{2}$ units (D) 4 units

Q2. Compare the following pairs of rational numbers using $<$, $=$ or $>$.

- (1) $3/5$ and $4/7$
- (2) $-5/6$ and $-7/9$
- (3) $7/8$ and $21/24$
- (4) $-2/3$ and $1/5$
- (5) $-11/12$ and $-13/15$

Q3. Write the following rational numbers in decimal form.

- (1) $5/8$
- (2) $7/12$
- (3) $-13/6$
- (4) $4/11$
- (5) $17/20$

Q4. State whether the following statements are True or False.

- (1) Every rational number is a real number.
- (2) π is exactly equal to $22/7$.
- (3) $3\sqrt{2}$ is an irrational number.
- (4) The decimal form of an irrational number is terminating.
- (5) $\sqrt{4}$ is an irrational number.

Q5. Show the following numbers on a number line (draw a separate number line for each).

- (1) $5/3$
- (2) $-3/4$
- (3) $-7/2$

Q6. Show $\sqrt{3}$ and $\sqrt{5}$ on a number line. Write the steps of construction.

Q7. Write any two rational numbers between $\frac{1}{3}$ and $\frac{1}{2}$.

Chapter 2 : Parallel Lines and Transversal

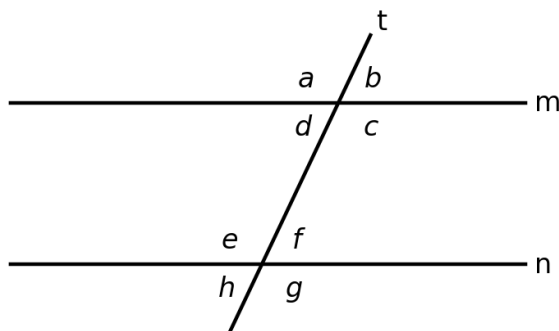


Figure 1 (line $m \parallel$ line n , t is a transversal)

Q1. Use Figure 1 (line $m \parallel$ line n , t is a transversal) and fill in the blanks.

- (1) The corresponding angle of $\angle a$ is _____.
- (2) The corresponding angle of $\angle c$ is _____.
- (3) The interior alternate angle of $\angle d$ is _____.
- (4) The interior alternate angle of $\angle c$ is _____.
- (5) The exterior alternate angle of $\angle a$ is _____.
- (6) $\angle d$ and $\angle e$ form a pair of _____ angles.

Q2. In Figure 1, line $m \parallel$ line n and $m\angle b = 110^\circ$. Find the measures of the remaining seven angles. Give reasons.

Q3. In Figure 1, line $m \parallel$ line n . If $m\angle a = (3x + 10)^\circ$ and $m\angle e = (2x + 30)^\circ$, find x and the measures of $\angle a$ and $\angle d$.

Q4. In Figure 1, line $m \parallel$ line n . If $m\angle d = (3x + 20)^\circ$ and $m\angle e = (2x + 10)^\circ$, find x and the measure of each angle.

Q5. Choose the correct alternative.

- (1) When two parallel lines are cut by a transversal, the number of angles formed is
(A) 4 (B) 6 (C) 8 (D) 12
- (2) Each pair of interior angles formed by two parallel lines and a transversal is
(A) congruent (B) supplementary (C) complementary (D) equal to 90°
- (3) One of the interior alternate angles formed by two parallel lines and a transversal is 65° . The other is
(A) 115° (B) 25° (C) 65° (D) 90°

Q6. Two lines are cut by a transversal. Decide with reason whether the lines are parallel.

- (1) A pair of corresponding angles measures 60° and 60° .
- (2) A pair of interior angles measures 100° and 70° .
- (3) A pair of alternate angles measures 45° and 55° .

Q7. Construction.

- (1) Draw a line l and take a point P outside it. Draw a line through P parallel to l using two set-squares.
- (2) Draw a line m . Draw a line n parallel to m at a distance of 3.5 cm from it.

Chapter 3 : Indices and Cube Root

Q1. Express the following in index form.

- (1) Fifth root of 13
- (2) Sixth root of 9
- (3) Square root of 256
- (4) Cube root of 17
- (5) Seventh root of 30

Q2. Write each of the following in the form 'nth root of a'.

- (1) $7^{1/3}$
- (2) $15^{1/5}$
- (3) $128^{1/7}$
- (4) $900^{1/12}$
- (5) $(11)^{1/4}$

Q3. Find the value.

- (1) $25^{1/2}$
- (2) $64^{1/3}$
- (3) $8^{2/3}$
- (4) $243^{1/5}$
- (5) $27^{2/3}$
- (6) $16^{3/4}$

Q4. Fill in the blanks using laws of indices.

- (1) $3^5 \times 3^2 = 3$ —
- (2) $3^7 \div 3^9 = 3$ —
- (3) $(3^4)^5 = 3$ —
- (4) $5^{-3} = 1/\underline{\hspace{1cm}}$
- (5) $(2/3)^{-2} = \underline{\hspace{1cm}}$

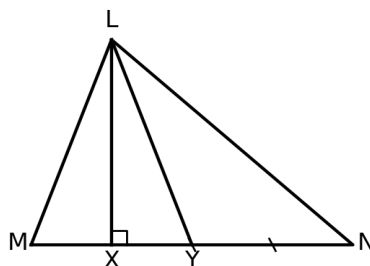
Q5. Simplify.

- (1) $2^{1/2} \times 2^{1/2}$
 (2) $9^{1/2} \times 27^{1/3}$
 (3) $(5^{1/3})^3$
 (4) $7^{2/3} \times 7^{1/3}$

Q6. Choose the correct alternative.

- (1) The value of $4^{3/2}$ is
(A) 6 (B) 8 (C) 12 (D) 64
- (2) If $x^{1/3} = 4$, then $x =$
(A) 12 (B) 16 (C) 64 (D) 81
- (3) If $10^{1/5} = x$, then
(A) $x^5 = 10$ (B) $x^{10} = 5$ (C) $5x = 10$ (D) $x = 50$

Chapter 4 : Altitudes and Medians of a Triangle



Q1. In Figure 2, $\triangle LMN$ is shown. Fill in the blanks.

- (1) The altitude of $\triangle LMN$ on side MN is seg _____.
- (2) The median of $\triangle LMN$ on side MN is seg _____.
- (3) Y is the _____ of side MN .

Q2. Choose the correct alternative.

- (1) The point of concurrence of the three altitudes of a triangle is called
(A) centroid (B) orthocentre (C) incentre (D) circumcentre
- (2) The orthocentre of a right-angled triangle is
(A) inside the triangle (B) outside the triangle (C) the vertex of the right angle
(D) the midpoint of the hypotenuse
- (3) The centroid divides each median in the ratio
(A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 3 : 1
- (4) The orthocentre of an obtuse-angled triangle lies
(A) in the interior (B) in the exterior (C) on a side (D) at a vertex

Q3. G is the centroid of $\triangle ABC$. Seg AP , seg BQ and seg CR are its medians.

- (1) If $I(AG) = 8$ cm, find $I(GP)$ and $I(AP)$.
- (2) If $I(BG) = 10$ cm, find $I(GQ)$ and $I(BQ)$.
- (3) If $I(GR) = 3$ cm, find $I(CG)$ and $I(CR)$.

Q4. Constructions.

- (1) Draw an acute-angled $\triangle PQR$ with sides 5 cm, 6 cm and 7 cm. Draw all its altitudes and name the point of concurrence O .
- (2) Draw an obtuse-angled $\triangle STV$ with $I(ST) = 4$ cm, $I(TV) = 5$ cm and $I(SV) = 8$ cm. Draw its medians and show the centroid G .
- (3) Draw a right-angled $\triangle XYZ$ with $\angle Y = 90^\circ$. Draw its altitudes and mark the orthocentre.
- (4) Draw an isosceles triangle. Draw all its medians and altitudes. Write your observation.

Q5. Complete the table.

Type of triangle	Position of orthocentre	Position of centroid
Acute-angled		
Right-angled		
Obtuse-angled		

Q6. Write the difference between an altitude and a median of a triangle.

Q7. Draw an equilateral triangle. Locate its circumcentre, incentre, centroid and orthocentre. Write your observation.

Chapter 5 : Expansion Formulae

Q1. Choose the correct alternative.

- (1) $(x + 3)(x - 2) =$
(A) $x^2 + x - 6$ (B) $x^2 - x - 6$ (C) $x^2 + 5x - 6$ (D) $x^2 + x + 6$
- (2) The coefficient of x in the expansion of $(x + 7)(x - 3)$ is
(A) 10 (B) 4 (C) -4 (D) -21
- (3) $(a + b)^3 =$
(A) $a^3 + b^3$ (B) $a^3 + b^3 + 3ab(a + b)$ (C) $a^3 + b^3 - 3ab(a + b)$ (D) $a^3 + 3a^2b + b^3$
- (4) $(a + b + c)^2 =$
(A) $a^2 + b^2 + c^2$ (B) $a^2 + b^2 + c^2 + ab + bc + ca$ (C) $a^2 + b^2 + c^2 + 2(ab + bc + ca)$
(D) $a^2 + b^2 + c^2 - 2(ab + bc + ca)$

Q2. Expand.

- (1) $(x + 5)(x + 7)$
(2) $(y + 9)(y - 4)$
(3) $(m - 6)(m - 8)$
(4) $(2a + 3b)(2a - 5b)$
(5) $(x + 1/3)(x + 1/2)$

Q3. Expand using the formula for the cube of a binomial.

- (1) $(x + 4)^3$
(2) $(2m + 3)^3$
(3) $(3p - 2)^3$
(4) $(5 - x)^3$
(5) $(2x - 3y)^3$

Q4. Find the value using suitable expansion formulae.

- (1) $(103)^2$
(2) $(97)^2$
(3) 102×98
(4) $(101)^3$
(5) $(99)^3$

Q5. Simplify.

- (1) $(x + y)^3 - (x - y)^3$
(2) $(2a + b)^3 + (2a - b)^3$

Q6. Expand.

- (1) $(x + y + 2)^2$
(2) $(3x + 2y - z)^2$

Q7. Solve.

- (1) If $a + b = 7$ and $ab = 10$, find $a^3 + b^3$.
(2) If $a - b = 3$ and $ab = 10$, find $a^3 - b^3$.
(3) If $x + 1/x = 3$, find $x^3 + 1/x^3$.

Chapter 6 : Factorisation of Algebraic Expressions

Q1. Choose the correct alternative.

- (1) The factors of $x^2 - 7x + 12$ are
(A) $(x - 3)(x - 4)$ (B) $(x + 3)(x + 4)$ (C) $(x - 2)(x - 6)$ (D) $(x + 3)(x - 4)$
- (2) $a^3 + b^3 =$
(A) $(a + b)(a^2 + ab + b^2)$ (B) $(a + b)(a^2 - ab + b^2)$ (C) $(a - b)(a^2 + ab + b^2)$ (D) $(a - b)(a^2 - ab + b^2)$
- (3) $a^3 - b^3 =$
(A) $(a - b)(a^2 + ab + b^2)$ (B) $(a - b)(a^2 - ab + b^2)$ (C) $(a + b)(a^2 - ab + b^2)$ (D) $(a - b)^3$

Q2. Factorise the following quadratic trinomials.

- (1) $x^2 + 8x + 15$
(2) $x^2 - 9x + 20$
(3) $p^2 + 2p - 24$
(4) $2x^2 + 7x + 3$
(5) $3y^2 - 10y + 8$
(6) $6x^2 - x - 12$
(7) $3x^2 - 12x - 36$

Q3. Factorise using the formula for $a^3 + b^3$.

- (1) $x^3 + 8$
(2) $27a^3 + 64b^3$
(3) $54p^3 + 250q^3$
(4) $y^3 + 1/27$

Q4. Factorise using the formula for $a^3 - b^3$.

- (1) $y^3 - 125$
(2) $8x^3 - 27y^3$
(3) $128a^3 - 2b^3$
(4) $1 - 64m^3$

Q5. Simplify.

- (1) $(a + b)^3 - (a^3 + b^3)$
(2) $(x^2 - 9) / (x^2 + 6x + 9)$
(3) $(x^3 - 27) / (x^2 - 9)$
(4) $(x^2 - 5x + 6) / (x^2 - 4) \times (x + 2) / (x - 3)$
(5) $(x^2 - 4) / (x^2 + x - 6) \div (x + 2) / (x^2 - 9)$

Chapter 7 : Variation

Q1. Write the following statements using the symbol of variation ($\propto$).

- (1) The area (A) of a circle is directly proportional to the square of its radius (r).
(2) At a constant speed, the distance (d) covered by a car varies directly as the time (t).
(3) At a constant distance, the speed (s) of a vehicle varies inversely as the time (t) taken.
(4) The intensity (I) of light varies inversely as the square of the distance (d) from the lamp.

Q2. Solve the following.

- (1) x varies directly as y. When $x = 12$, $y = 4$. Find the constant of variation, the equation of variation and the value of x when $y = 10$.

- (2) y varies directly as the square of x . When $x = 4$, $y = 48$. Find y when $x = 6$.
- (3) y varies directly as the square root of x . When $x = 25$, $y = 10$. Find y when $x = 64$.
- (4) a varies inversely as b . When $a = 8$, $b = 6$. Find the constant of variation. Find b when $a = 12$ and a when $b = 16$.
- (5) $f \propto 1/d^2$. When $d = 4$, $f = 25$. Find f when $d = 10$, and d when $f = 16$.

Q3. The number of pencils and their cost are in direct variation. Complete the table.

Number of pencils (n)	2	5	8	
Cost in ₹ (c)	14			84

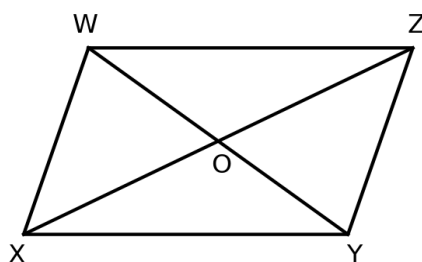
Q4. Solve the following word problems.

- (1) The cost of 5 kg of apples is ₹ 600. Find the cost of 12 kg of apples.
- (2) 12 workers can finish a job in 15 days. How many days will 20 workers take to finish the same job?
- (3) A vehicle running at 60 km/hr takes 5 hours to complete a journey. How much time will it take at 75 km/hr?
- (4) 8 pipes of the same size fill a tank in 6 hours. In how much time will 12 such pipes fill the tank?
- (5) The remuneration of 6 labourers is ₹ 1800. Find the remuneration of 15 labourers, if remuneration varies directly as the number of labourers.

Q5. Which of the following are examples of inverse variation? Which are of direct variation?

- (1) Number of workers and time taken to complete a job.
- (2) Quantity of petrol filled in a vehicle and its cost.
- (3) Speed of a vehicle and time taken to cover a fixed distance.
- (4) Number of books and their total cost.

Chapter 8 : Quadrilateral – Constructions and Types



Q1. Choose the correct alternative.

- (1) The diagonals of a rhombus
 (A) are equal only (B) bisect each other at right angles (C) are parallel (D) do not intersect
- (2) Which quadrilateral has exactly one pair of opposite sides parallel?
 (A) Parallelogram (B) Rhombus (C) Trapezium (D) Rectangle
- (3) Adjacent angles of a parallelogram are
 (A) congruent (B) complementary (C) supplementary (D) each 90°
- (4) In a kite,
 (A) all sides are equal (B) two pairs of adjacent sides are congruent (C) all angles are equal (D) diagonals are equal

Q2. Figure 3 shows parallelogram WXYZ. Its diagonals intersect at O. Write the answers.

- (1) If $l(WZ) = 6.4$ cm, then $l(XY) = ?$
- (2) If $l(YZ) = 5.2$ cm, then $l(XW) = ?$
- (3) If $l(OX) = 3.1$ cm, then $l(OZ) = ?$
- (4) If $l(WO) = 4.2$ cm, then $l(WY) = ?$
- (5) If $m\angle WZY = 115^\circ$, then $m\angle WXY = ?$ and $m\angle XWZ = ?$

Q3. Solve.

- (1) The diagonal of a rectangle is 13 cm and one side is 5 cm. Find the other side.
- (2) Find the length of the diagonal of a square of side 9 cm.
- (3) The diagonals of a rhombus are 24 cm and 10 cm. Find its side and perimeter.
- (4) One angle of a rhombus measures 65° . Find the measures of the remaining three angles.
- (5) The adjacent angles of a parallelogram are $(3x - 5)^\circ$ and $(2x + 15)^\circ$. Find the measure of each angle.
- (6) The angles of a quadrilateral are in the ratio 3 : 4 : 5 : 6. Find the measure of each angle and write, with reason, the type of quadrilateral.

Q4. State whether the following statements are True or False.

- (1) The diagonals of a rhombus are equal.
- (2) The diagonals of a rectangle bisect each other.
- (3) Every square is a rhombus.
- (4) Every trapezium is a parallelogram.
- (5) In a kite, one diagonal is the perpendicular bisector of the other.

Q5. Constructions.

- (1) Construct $\square ABCD$ such that $l(AB) = 4$ cm, $l(BC) = 5$ cm, $l(CD) = 4.5$ cm, $l(DA) = 3.5$ cm and $l(AC) = 6$ cm.
- (2) Construct $\square PQRS$ such that $l(PQ) = 4.5$ cm, $l(QR) = 5.5$ cm, $m\angle P = 70^\circ$, $m\angle Q = 100^\circ$ and $m\angle R = 90^\circ$.
- (3) Construct a rectangle ABCD with $l(AB) = 7$ cm and $l(BC) = 4$ cm.
- (4) Construct a rhombus KLMN with side 4.5 cm and $m\angle K = 60^\circ$.
- (5) Construct a parallelogram ABCD such that $l(BC) = 7$ cm, $m\angle ABC = 40^\circ$ and $l(AB) = 3$ cm.

Q6. Write any two properties each of a rectangle, a square, a rhombus and a parallelogram.

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Worksheet – Answer Key

Chapter 1 : Rational and Irrational Numbers

- Q1. (1) B (2) B (3) A (4) B (5) A
- Q2. (1) $\frac{3}{5} > \frac{4}{7}$ [$3 \times 7 = 21 > 5 \times 4 = 20$] (2) $-\frac{5}{6} < -\frac{7}{9}$ [$\frac{5}{6} = \frac{15}{18}$ and $\frac{7}{9} = \frac{14}{18}$, so $\frac{5}{6} > \frac{7}{9}$] (3) $\frac{7}{8} = \frac{21}{24}$ (4) $-\frac{2}{3} < \frac{1}{5}$ [negative < positive] (5) $-\frac{11}{12} < -\frac{13}{15}$ [$\frac{11}{12} = \frac{55}{60} > \frac{13}{15} = \frac{52}{60}$]
- Q3. (1) 0.625 (2) $0.58333... = 0.58\overline{3}$ (3 recurring) (3) $-2.1666... = -2.1\overline{6}$ (6 recurring) (4) $0.3636... = 0.3\overline{6}$ (36 recurring) (5) 0.85
- Q4. (1) True (2) False ($\frac{22}{7}$ is only an approximation) (3) True (4) False (it is non-terminating, non-recurring) (5) False ($\sqrt{4} = 2$, which is rational)
- Q5. (1) Divide each unit into 3 equal parts; $\frac{5}{3} = 1 + \frac{2}{3}$, so mark the 5th point to the right of 0. (2) Divide each unit into 4 parts; mark the 3rd point to the left of 0. (3) $-\frac{7}{2} = -3\frac{1}{2}$; mark the midpoint between -3 and -4.
- Q6. $\sqrt{3}$: On the number line take $OQ = \sqrt{2}$ (as in Q1 of Practice Set 1.4). Draw $QR \perp OQ$ with $QR = 1$ unit. Join OR : $l(OR)^2 = (\sqrt{2})^2 + 1^2 = 3$, so $l(OR) = \sqrt{3}$. With centre O and radius OR draw an arc cutting the number line at C ; C represents $\sqrt{3}$. $\sqrt{5}$: Draw a right triangle with perpendicular sides 2 units and 1 unit; the hypotenuse is $\sqrt{5}$. Draw an arc with centre O and this radius to cut the number line.
- Q7. Any two, e.g. $\frac{3}{8}$ (0.375) and $\frac{5}{12}$ (0.4166...). [$\frac{1}{3} = 0.333...$, $\frac{1}{2} = 0.5$]

Chapter 2 : Parallel Lines and Transversal

- Q1. (1) $\angle e$ (2) $\angle g$ (3) $\angle f$ (4) $\angle e$ (5) $\angle g$ (6) interior
- Q2. $\angle a = 70^\circ$ (linear pair with $\angle b$); $\angle c = 70^\circ$ (vertical to $\angle a$); $\angle d = 110^\circ$ (vertical to $\angle b$); $\angle f = 110^\circ$ (corresponding to $\angle b$); $\angle e = 70^\circ$ (corresponding to $\angle a$); $\angle g = 70^\circ$ (corresponding to $\angle c$); $\angle h = 110^\circ$ (corresponding to $\angle d$).
- Q3. $\angle a$ and $\angle e$ are corresponding angles, so $3x + 10 = 2x + 30$, $x = 20$. $m\angle a = 3(20) + 10 = 70^\circ$. $m\angle d = 180^\circ - 70^\circ = 110^\circ$ (linear pair).
- Q4. $\angle d$ and $\angle e$ are interior angles, hence supplementary: $(3x + 20) + (2x + 10) = 180$, $5x = 150$, $x = 30$. $m\angle d = 110^\circ$, $m\angle e = 70^\circ$.
- Q5. (1) C (2) B (3) C
- Q6. (1) Parallel (corresponding angles are congruent). (2) Not parallel ($100^\circ + 70^\circ = 170^\circ \neq 180^\circ$). (3) Not parallel (alternate angles $45^\circ \neq 55^\circ$).
- Q7. (1) Draw line l and point P . Place two set-squares touching each other, keep one fixed, slide the other until its edge passes through P , and draw line m along that edge. $m \parallel l$. (2) Draw line m , take two points A and B on it, draw perpendiculars at A and B , mark P and Q on them at 3.5 cm from A and B , draw line PQ . $PQ \parallel m$ at 3.5 cm distance.

Chapter 3 : Indices and Cube Root

- Q1. (1) $13^{1/5}$ (2) $9^{1/6}$ (3) $256^{1/2}$ (4) $17^{1/3}$ (5) $30^{1/7}$
- Q2. (1) cube root of 7 (2) 5th root of 15 (3) 7th root of 128 (4) 12th root of 900 (5) 4th root of 11
- Q3. (1) 5 (2) 4 (3) 4 (4) 3 (5) 9 (6) 8
- Q4. (1) 3^7 (2) 3^{-2} (3) 3^{20} (4) $\frac{1}{5^3} = \frac{1}{125}$ (5) $\frac{9}{4}$
- Q5. (1) 2 (2) $3 \times 3 = 9$ (3) 5 (4) $7^1 = 7$
- Q6. (1) B ($4^{3/2} = (\sqrt{4})^3 = 8$) (2) C (3) A

Chapter 4 : Altitudes and Medians of a Triangle

- Q1. (1) LX (2) LY (3) midpoint
- Q2. (1) B (2) C (3) C (4) B

Q3. (1) $l(GP) = 4$ cm, $l(AP) = 12$ cm [$AG : GP = 2 : 1$] (2) $l(GQ) = 5$ cm, $l(BQ) = 15$ cm (3) $l(CG) = 6$ cm, $l(CR) = 9$ cm

Q4. (1) Construct the triangle from the three sides; draw a perpendicular from each vertex to the opposite side with a set-square. The three altitudes meet at O inside the triangle. (2) Construct the triangle; find the midpoint of each side and join it to the opposite vertex; the medians meet at G inside the triangle. (3) The altitudes meet at vertex Y, the right-angle vertex (orthocentre = Y). (4) In an isosceles triangle the median and altitude from the vertex angle coincide; the centroid and orthocentre lie on this line.

Q5. Acute-angled: orthocentre inside, centroid inside. Right-angled: orthocentre at the vertex of the right angle, centroid inside. Obtuse-angled: orthocentre outside, centroid inside.

Q6. Altitude: perpendicular segment from a vertex to the opposite side. Median: segment joining a vertex to the midpoint of the opposite side. Altitudes meet at the orthocentre; medians meet at the centroid, which divides each median 2 : 1.

Q7. In an equilateral triangle the circumcentre, incentre, centroid and orthocentre coincide (all four are the same point).

Chapter 5 : Expansion Formulae

Q1. (1) A (2) B (3) B (4) C

Q2. (1) $x^2 + 12x + 35$ (2) $y^2 + 5y - 36$ (3) $m^2 - 14m + 48$ (4) $4a^2 - 4ab - 15b^2$ (5) $x^2 + (5/6)x + 1/6$

Q3. (1) $x^3 + 12x^2 + 48x + 64$ (2) $8m^3 + 36m^2 + 54m + 27$ (3) $27p^3 - 54p^2 + 36p - 8$ (4) $125 - 75x + 15x^2 - x^3$ (5) $8x^3 - 36x^2y + 54xy^2 - 27y^3$

Q4. (1) $(100 + 3)^2 = 10609$ (2) $(100 - 3)^2 = 9409$ (3) $(100 + 2)(100 - 2) = 9996$ (4) $(100 + 1)^3 = 1030301$ (5) $(100 - 1)^3 = 970299$

Q5. (1) $6x^2y + 2y^3$ (2) $16a^3 + 12ab^2$

Q6. (1) $x^2 + y^2 + 4 + 2xy + 4y + 4x$ (2) $9x^2 + 4y^2 + z^2 + 12xy - 4yz - 6xz$

Q7. (1) $a^3 + b^3 = (a + b)^3 - 3ab(a + b) = 343 - 210 = 133$ (2) $a^3 - b^3 = (a - b)^3 + 3ab(a - b) = 27 + 90 = 117$ (3) $x^3 + 1/x^3 = (x + 1/x)^3 - 3(x + 1/x) = 27 - 9 = 18$

Chapter 6 : Factorisation of Algebraic Expressions

Q1. (1) A (2) B (3) A

Q2. (1) $(x + 3)(x + 5)$ (2) $(x - 4)(x - 5)$ (3) $(p + 6)(p - 4)$ (4) $(2x + 1)(x + 3)$ (5) $(3y - 4)(y - 2)$ (6) $(2x - 3)(3x + 4)$ (7) $3(x - 6)(x + 2)$

Q3. (1) $(x + 2)(x^2 - 2x + 4)$ (2) $(3a + 4b)(9a^2 - 12ab + 16b^2)$ (3) $2(3p + 5q)(9p^2 - 15pq + 25q^2)$ (4) $(y + 1/3)(y^2 - y/3 + 1/9)$

Q4. (1) $(y - 5)(y^2 + 5y + 25)$ (2) $(2x - 3y)(4x^2 + 6xy + 9y^2)$ (3) $2(4a - b)(16a^2 + 4ab + b^2)$ (4) $(1 - 4m)(1 + 4m + 16m^2)$

Q5. (1) $3a^2b + 3ab^2 = 3ab(a + b)$ (2) $(x - 3)/(x + 3)$ (3) $(x^2 + 3x + 9)/(x + 3)$ (4) 1 (5) $x - 3$

Chapter 7 : Variation

Q1. (1) $A \propto r^2$ (2) $d \propto t$ (3) $s \propto 1/t$ (4) $l \propto 1/d^2$

Q2. (1) $k = 12/4 = 3$; equation $x = 3y$; when $y = 10$, $x = 30$. (2) $y = kx^2$: $48 = 16k$, $k = 3$; $y = 3 \times 36 = 108$. (3) $y = k\sqrt{x}$: $10 = 5k$, $k = 2$; $y = 2 \times 8 = 16$. (4) $a \times b = k$: $k = 48$; $b = 48/12 = 4$; $a = 48/16 = 3$. (5) $f \times d^2 = k$: $k = 25 \times 16 = 400$; $f = 400/100 = 4$; $d^2 = 400/16 = 25$, $d = 5$.

Q3. $c = 7n$. $n = 5 \rightarrow ₹ 35$; $n = 8 \rightarrow ₹ 56$; $c = 84 \rightarrow n = 12$.

Q4. (1) ₹ 120 per kg; cost of 12 kg = ₹ 1440. (2) workers $\times$ days = $12 \times 15 = 180$; days = $180/20 = 9$ days. (3) speed $\times$ time = 300; time = $300/75 = 4$ hours. (4) pipes $\times$ time = 48; time = $48/12 = 4$ hours. (5) ₹ 300 per labourer; remuneration = ₹ 4500.

Q5. Inverse variation: (1) and (3). Direct variation: (2) and (4).

Chapter 8 : Quadrilateral – Constructions and Types

Q1. (1) B (2) C (3) C (4) B

Q2. (1) 6.4 cm (2) 5.2 cm (3) 3.1 cm (4) 8.4 cm (5) $m\angle WXY = 115^\circ$, $m\angle XWZ = 65^\circ$

Q3. (1) Other side = $\sqrt{13^2 - 5^2} = \sqrt{144} = 12$ cm. (2) Diagonal = $\sqrt{9^2 + 9^2} = \sqrt{162} = 9\sqrt{2}$ cm. (3) Half-diagonals are 12 cm and 5 cm; side = $\sqrt{144 + 25} = 13$ cm; perimeter = 52 cm. (4) $115^\circ, 65^\circ, 115^\circ$. (5) $(3x - 5) + (2x + 15) = 180$, $5x = 170$, $x = 34$; angles 97° and 83° (and $97^\circ, 83^\circ$ for the other pair). (6) $3x + 4x + 5x + 6x = 360$, $x = 20$; angles $60^\circ, 80^\circ, 100^\circ, 120^\circ$. Since $60^\circ + 120^\circ = 180^\circ$ and $80^\circ + 100^\circ = 180^\circ$, one pair of opposite sides is parallel, while $60^\circ + 80^\circ \neq 180^\circ$. So it is a trapezium.

Q4. (1) False (2) True (3) True (4) False (5) True

Q5. (1) Draw $AC = 6$ cm. Construct $\triangle ABC$ with $AB = 4$ cm, $BC = 5$ cm on one side and $\triangle ADC$ with $AD = 3.5$ cm, $CD = 4.5$ cm on the other side of AC . Join to complete $ABCD$. (2) Draw $PQ = 4.5$ cm; make $\angle Q = 100^\circ$ and $QR = 5.5$ cm; at P make $\angle P = 70^\circ$; at R make $\angle R = 90^\circ$; the rays meet at S . (3) Draw $AB = 7$ cm, draw perpendiculars at A and B , cut off 4 cm on each, join to get the rectangle. (4) Draw $KL = 4.5$ cm, $\angle K = 60^\circ$, $KN = 4.5$ cm; with centres N and L and radius 4.5 cm draw arcs meeting at M ; join NM and LM . (5) Draw $BC = 7$ cm, $\angle ABC = 40^\circ$ with $AB = 3$ cm; draw the line through A parallel to BC and through C parallel to BA ; they meet at D .

Q6. Rectangle: opposite sides congruent; diagonals congruent and bisect each other. Square: all sides congruent; diagonals are equal, bisect each other at 90° and bisect the opposite angles. Rhombus: opposite angles congruent; diagonals bisect each other at right angles and bisect the angles. Parallelogram: opposite sides and opposite angles congruent; diagonals bisect each other; adjacent angles are supplementary.